

KEY LEMMAS (for verification)

MO 413935 · limit OPEN · three load-bearing statements

SETUP

$$m_n = \min_A \max_x |Q_A(x)|, \quad Q_A(x) = \sum_{i < j} A_{ij} x_i x_j, \quad \alpha_n = m_n / n^{3/2}.$$

Question: does $\lim \alpha_n$ exist? Identifying a value is separate.

Reviewed sandwich: $B \leq \liminf \alpha_n \leq \limsup \alpha_n \leq 1/2$,

$B = 0.3258407554... > 13/40$. Convergence is not implied.

LEMMA A — paired-field lower bound (2026-09-17)

Split polynomial-phase sign covariances $C_s = C_0 + s \cdot 2\kappa M / (1 + q) + E_s$,

$\kappa = 2/\pi$, $\|E_s\|_F = O_L(1)$. Retain the PSD even tail.

Pair local variances by $(M^3)_{ij}^2 \leq q[(M^4)_{ii} - q^2]$.

Cap L is removed by existing same-order regularization ($n \rightarrow \infty$ first).

Then $\liminf \alpha_n \geq B = [81\kappa/2 + 9\sqrt{\kappa(2/3 + \kappa/2)}]/101$.

This is an all-orders lower bound, not a finite cutoff and not a limit.

Nine exact scalar/polynomial checks passed (NUKA). Not a formalization.

CHECK

- $\kappa = 2/\pi \Rightarrow B \approx 0.3258407554 > 13/40 = 0.325$.
- Conference constructions still give only $\limsup \alpha_n \leq 1/2$.
- Proof: evidence/NOTE_2026-09-17_PAURED_POLYNOMIAL_FIELD_LOWER.md

KEY LEMMAS (continued)

two-ray · residue (6.20) · finite census

LEMMA B — two-ray criterion (CORE §7 / Prop 6.3)

Put $H(n) = m_n^{2/3}$. If a nonnegative η has Dini tail $E(N) \rightarrow 0$ and

$$H(2n) \leq 2H(n) + 2n\eta(n), \quad H(3n) \leq 3H(n) + 3n\eta(n)$$

for all large n , then α_n converges. Proof: $h = H/n$ changes by η along words in $\{2, 3\}$; $\log 2/\log 3$ irrational $\Rightarrow$ ratio-dense semigroup.

This is sufficient, not necessary. Neither amplification ray is proved.

Paley-skew shielding (Prop 6.6) reduces doubling to residue (6.20).

LEMMA C — residue (6.20) is live on exact minimizers

Exact $m_5 = 4$, $m_6 = 5$, $m_8 = 10$: residue occupies 55%, 91%, 51% of pairs.

On that set, Paley-skew R violates the diamond by 0.42, 0.53, 0.34 in units of $n^{3/2}$. High- α random signings have sparse residue and hold.

So closing multiplier two for actual minimizers cannot use the global Paley estimate (6.24). Coupling of Q_A with $x^T R y$ on a large set is required.

Finite only. Does not close or refute the all-orders ray.

ASSEMBLY

A + conference upper bound $\Rightarrow$ sandwich. B is a possible close, unused.

C shows the remaining doubling gate is not an empty finite artifact.

Original limit remains OPEN. Do not cite $L = 1/2$.